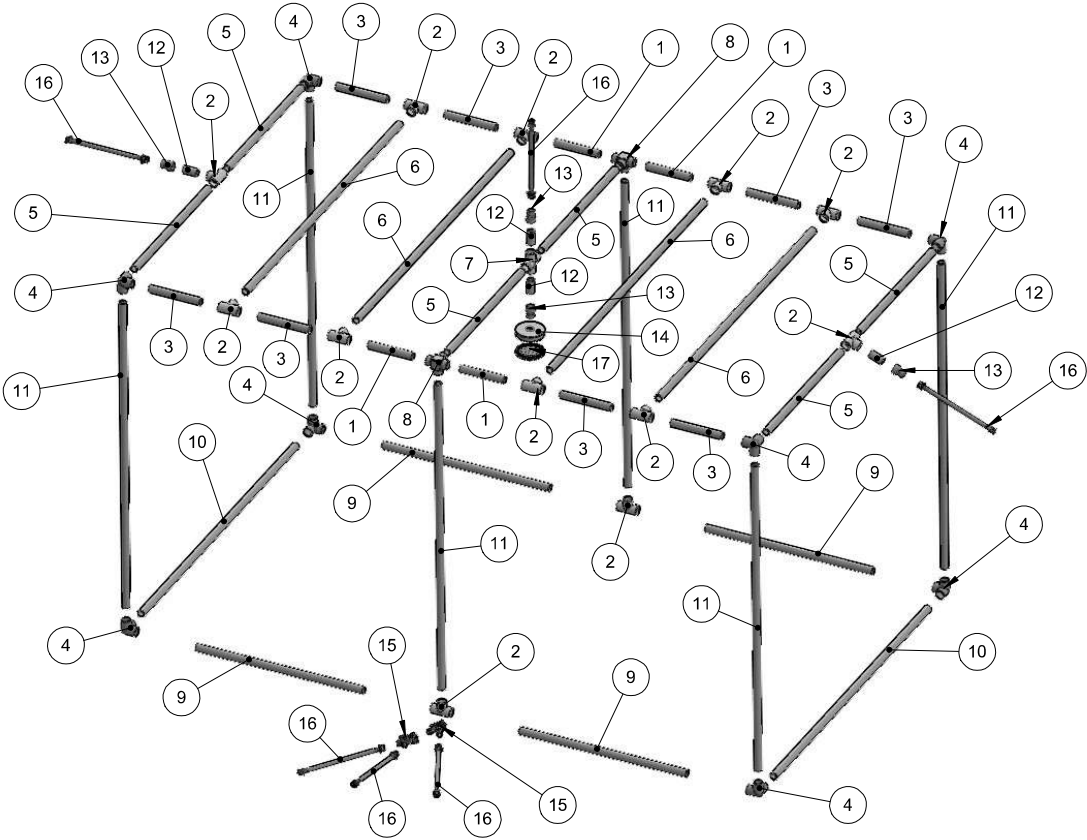


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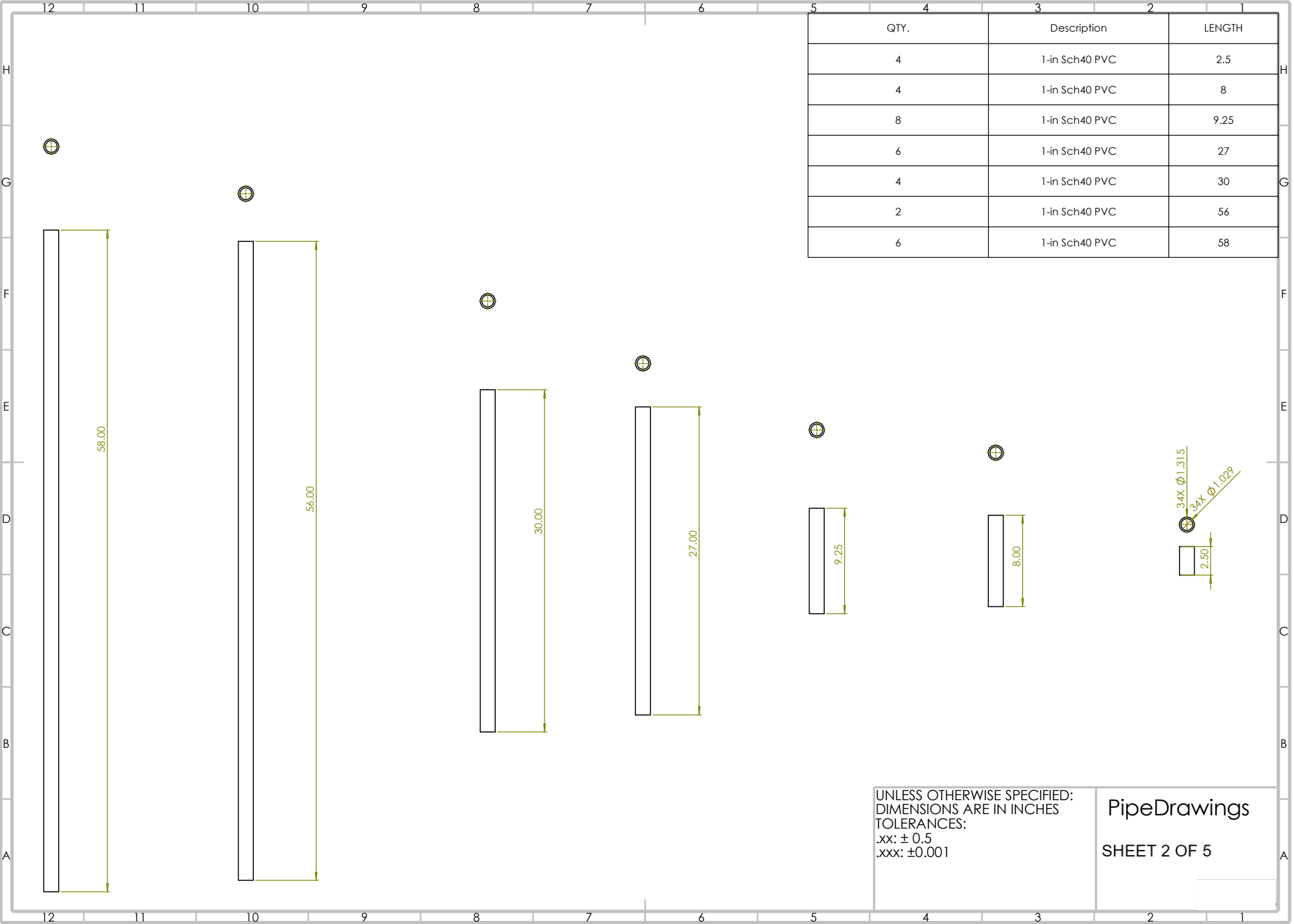
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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	8in PVC		4
2	1", TEE, PVC, SCH 40, SLP x SLP x SLP	1", TEE, PVC, SCH 40, SLP x SLP x SLP	12
3	9.25in PVC		8
4	4880K633	Standard-Wall PVC Pipe Fitting for Water	8
5	27in PVC		6
6	56in PVC with holes		4
7	1", CROSS, PVC, SCH 40, SLP	1", CROSS, PVC, SCH 40, SLP	1
8	ConnectorPartSAS		2
9	30in PVC		4
10	56in PVC		2
11	58in PVC		6
12	2.5in PVC		4
13	4880K682	Standard-Wall PVC Pipe Fitting for Water	4
14	headforRoundedGrate4		1
15	HoseSplitter	Wye-Shaped On/Off Valve	2
16	7454T31	Economical Garden Hose	6
17	roundedgrate3		1

REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	XXXXX	

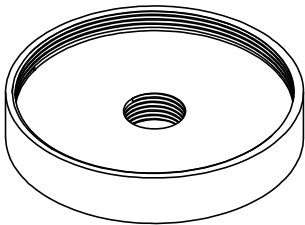
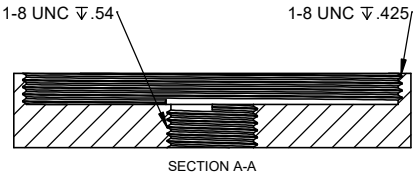
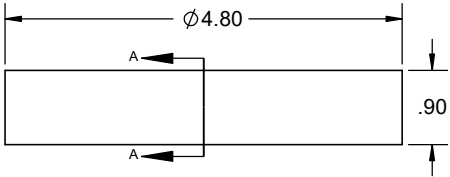
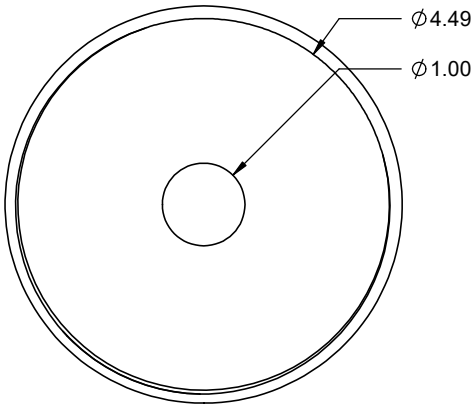


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	DIMENSIONS IN INCHES[MM]	MATERIAL	B		Compost Irrigation	
	DECIMAL: .xx = ±.010 .xxx = ±.005 .xxxx = ±.001	FINISH	DRAWING TITLE		Assembly	
	TYP EDGE BREAK: .005-.015 ANGULAR: ±0.5°	PROJECT CODE	DRAWN BY		CC	DATE
			DRAWING NUMBER			12/2/2025
			SHEET		1 OF 5	
			REV			



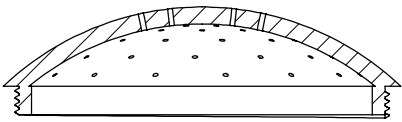
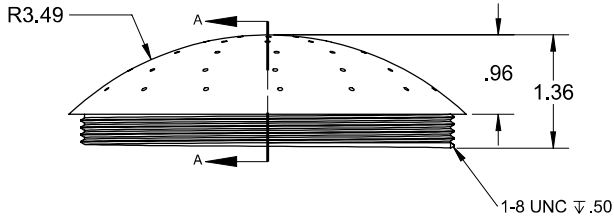
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REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	CC	12/1/25

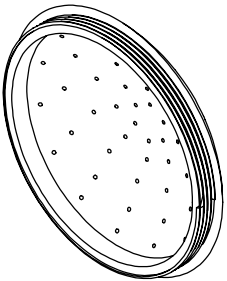
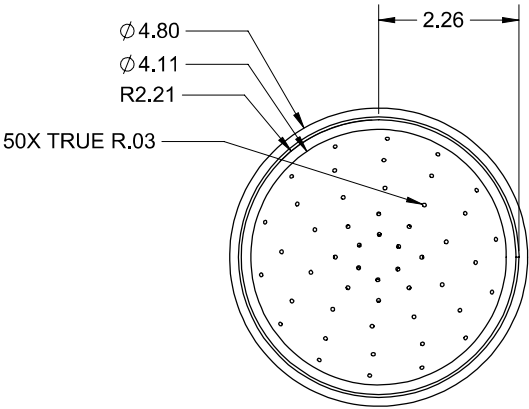


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	DIMENSIONS IN INCHES[MM]	MATERIAL	B	Compost Irrigation	
	DECIMAL: .xx = ±.015 .xxx = ±.005 .xxxx = ±.001	FINISH	DRAWING TITLE		
	TYP EDGE BREAK: .005-.015 ANGULAR: ±0.5°	PROJECT CODE	Showerhead Base		
		DRAWN BY		DATE	SHEET
		CC		12/1/2025	3 OF 5
		DRAWING NUMBER		REV	A

REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	CC	12/1/25



SECTION A-A



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TOLERANCES
(EXCEPT AS NOTED)
DIMENSIONS IN
INCHES[MM]

DECIMAL:
.xx = ±.015
.xxx = ±.005
.xxxx = ±.001

TYP EDGE BREAK:
.005-.015
ANGULAR:
±0.5°

NEXT ASSEMBLY

MATERIAL

FINISH

PROJECT CODE

SHEET SIZE

B

PROJECT

DRAWING TITLE

Showerhead Grate

DRAWN BY

CC

DATE

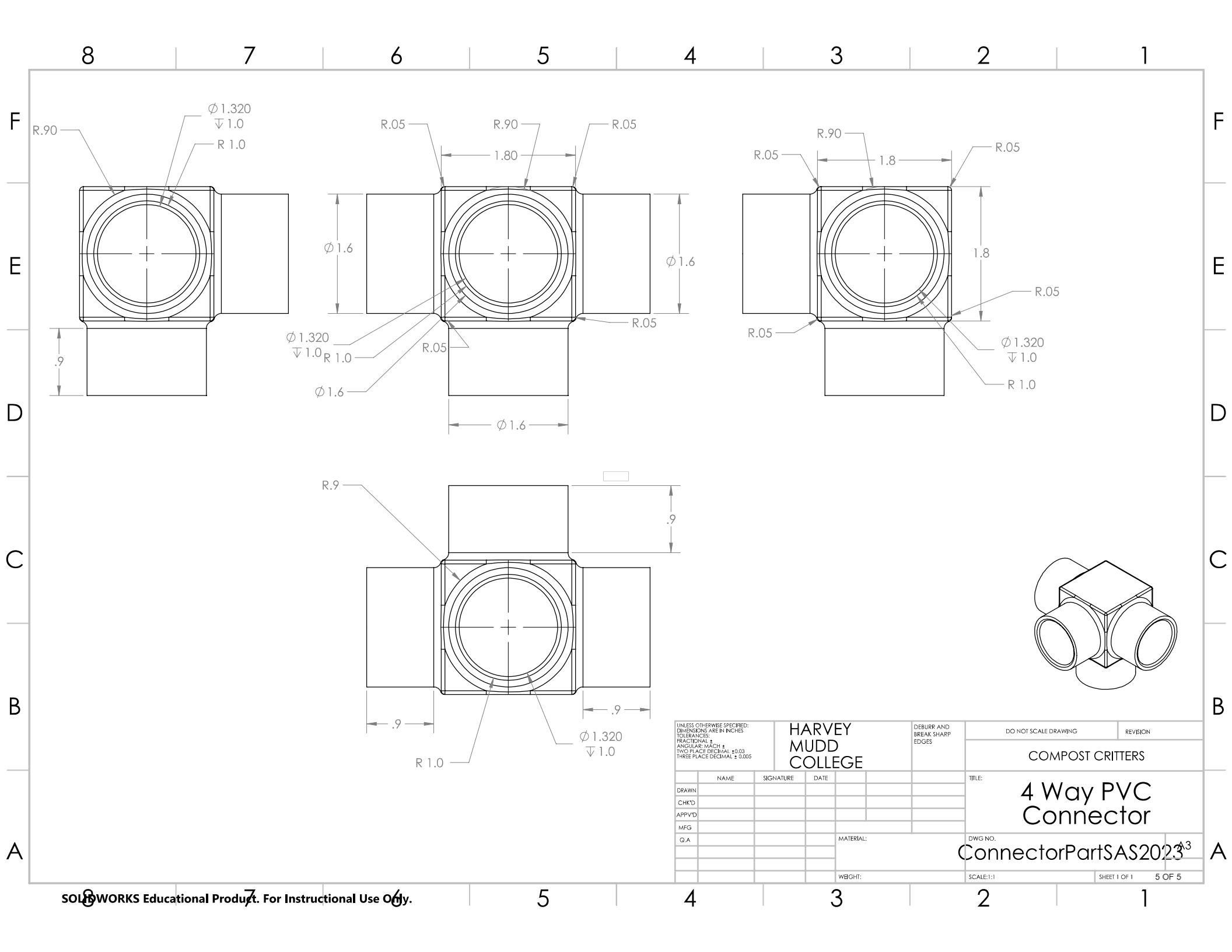
12/1/2025

SHEET

4 OF 5

REV

A



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL ± ANGULAR: MATCH ± TWO PLACE DECIMAL: ±0.03 THREE PLACE DECIMAL: ± 0.005				HARVEY MUDD COLLEGE			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING		REVISION	
COMPOST CRITTERS											
TITLE:										4 Way PVC Connector	
DRAWN	NAME	SIGNATURE	DATE								
CHK'D											
APP'D											
MFG											
G.A.				MATERIAL:			DWG NO.		13		
ConnectorPartSAS2023											
WEIGHT:							SCALE:1:1			SHEET 1 OF 1 5 OF 5	

Bill of Materials					
Item	Part Number	Manufacturer	Quantity	Price per unit (\$)	Total cost for part (\$)
20 ft 1-in Schedule 40 PVC pipe	23975	Charlotte Pipe	4	14.62	58.48
1-in Schedule 40 PVC Tee	23876	Charlotte Pipe	12	1.98	23.76
1-in Schedule 40 PVC Cross	22702	Charlotte Pipe	1	5.38	5.38
1-in Schedule 40 PVC Elbow	315499	Charlotte Pipe	8	5.22	41.76
3/4-in Plastic Threaded Drip Irrigation Female Adapter	5453992	Project Source	3	2.98	8.94
3/4-in x 1-in PVC Reducing Male Adapter	22695	Charlotte Pipe	3	1.78	5.34
Drip Irrigation Splitter	160885	Raindrip	2	3.18	6.36
5/8-in x 50-ft Drip Irrigation Distribution Tubing	62291	Raindrip	1	12.28	12.28
Universal Plastic Barb-Locking Collar Drip Irrigation Female Adapter	345804	Raindrip	2	2.58	5.16
Universal Plastic Barb-Locking Collar Drip Irrigation End Cap	340515	Raindrip	2	1.88	3.76
PLA 3d Printed 4-way PVC Connector			2	2	4
Resin 3d printed showerhead			1	18.72	18.72
				Total cost for prototype:	193.94

Compost Irrigator

Process	Machine/Tool	Comments
Cut PVC to the required lengths	PVC cutter, Tape measure	Cut 6x 58in, 6x 56in, 4x 30 in, 6x 27in, 8x 9.25in, 4x 8in 4x 2.5in
Deburr/smooth end	Deburring tool, sandpaper	Remove any loose parts using a deburring tool or sandpaper
Duct tape seal one end of PVC	Duct tape, Scissors	Duct tape one end of 4 of the 27in pieces (11), and 4 of the 8in pieces (part 1), cut around the edge of PVC to ensure PVC still fits into connector
Drill holes	Hand drill, 1/32in drill bit	Mark holes at 6in, 12in, 18in, 24in, 28in, 32in, 38in, 44in, and 50in from bottom of four of the 56in PVC pipes (part 6). Make sure to only drill one side of PVC.
Print Showerhead	Resin printer	Print in two parts
Print 2 PVC connecter pieces	PLA printer	Print two
Cut hose	Scissors	Cut three to about equal length
Connect PVC pieces		Assemble top frame first, then bottom frame, insert vertical pvc (part 11) into bottom frame, attach top frame to the vertical pvc (part 11).
Attach hose		Screw on hose splitter (part 15) to the other hose splitter (part 15). The free end will connect to the shower head with the other ends connecting to the side feeds.